

Analyzing Cardiovascular Diseases through Data Mining Techniques

----- Data Mining Final Project -----

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1. Background, Problem, and Motivation

1.1 Background and Motivation

According to the information of the World Health Organization (WHO), cardiovascular diseases (CVDs) take approximately 17.9 million lives each year. CVDs are a group of abnormal heart and blood vessel conditions, which include coronary heart disease, cerebrovascular disease, rheumatic heart disease, and others. Over 80% of cardiovascular disease (CVD) deaths result from heart attacks and strokes, with one-third of these fatalities occurring prematurely in individuals under the age of 70. It is worth noting that CVDs are one of the top 10 causes of death here in Taiwan. Based on the 2022 statistics of the Ministry of Health and Welfare, it is also among the top three leading causes of death.

Having unhealthy diets, physical inactivity, the usage of tobacco, and excessive alcohol consumption constitute the primary behavioral risk factors for heart disease and stroke. Individuals may exhibit the consequences of these behavioral risk factors as elevated blood pressure, increased blood glucose, heightened blood lipids, and overweight or obesity. These "intermediate risk factors" are measurable in healthcare facilities and signify risks of complications such as heart attack, stroke, and heart failure.

However, in most situations, there are no symptoms of the implicit disease of the blood vessels. The first indication of it may present itself as a heart attack or stroke, which may cause great damage to our body. Thus, analyzing cardiovascular diseases through data mining techniques may offer us more understanding of the relations among those intermediate risk factors.

1.2 Problem

As mentioned earlier, some intermediate risk factors contribute to CVDs. Examining the contribution of each factor and their relationships is more comprehensive than purely considering these factors once at a time. Using data mining techniques, the researchers can effortlessly extract valuable insights from large datasets and aid in decision-making. The following research addresses **‘what kinds of factors cause CVDs significantly and their relationships by constructing classification models’**.

2. Data Description and the Importance of Data

2.1. Data Description (Source)

The cardiovascular dataset in this final project proposal comes from Kaggle (<https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset/data>). The owner is Svetlana Ulianova, who uploaded it five years ago. There are no licenses and all the values in the dataset were collected at the moment of medical examination. However, there are few papers that seem to use the same dataset, which provides a great example to demonstrate the effectiveness of the dataset.

2.2.The Importance of Data

The dataset contains 70,000 records with 12 CVD-related attributes except for patients' IDs. In addition, there are 3 types of input features, which are *Objective*, *Examination*, and *Subjective*, respectively. Note that the objective feature means it is factual information, and the examination feature is the result of medical examinations, the subjective feature is the information given by the patient. Also, there are no missing values in the dataset. The detailed information is summarized in Table 1.

Table 1. Detailed information on the cardiovascular dataset

Detailed information on the cardiovascular dataset			
Column Name	Input Feature	Explanation	Data Type
id		Id number	integer
age	Objective Feature	Age	integer (days)
height	Objective Feature	Height	integer (cm)
weight	Objective Feature	Weight	integer (kg)
gender	Objective Feature	Gender	categorical code (1-women; 2-men)
ap_hi	Examination Feature	Systolic blood pressure	integer
ap_lo	Examination Feature	Diastolic blood pressure	integer
cholesterol	Examination Feature	Cholesterol	1: normal; 2: above normal; 3: well above normal
gluc	Examination Feature	Glucose	1: normal; 2: above normal; 3: well above normal
smoke	Subjective Feature	Smoking	binary
alco	Subjective Feature	Alcohol intake	binary
active	Subjective Feature	Physical activity	binary
cardio	Target Variable	Presence or absence of cardiovascular disease	binary

After analyzing the attributes, it was determined that 'id' holds no significance and will be removed in further processing. The features 'age', 'height', 'weight', 'ap_hi', and 'ap_lo' are numerical attributes, with the exception of 'age', which belongs to the interval scale. The remaining features are part of the ratio scale. The categorical parts consist of 'gender', 'cholesterol', 'gluc', 'smoke', 'alco', 'active', and 'cardio'. 'Cholesterol' and 'gluc' belong to the ordinal scale, while the others belong to the nominal scale. Notably, 'cardio' is the target value.

3. The Process Flow of Data Analysis and Data Preprocessing

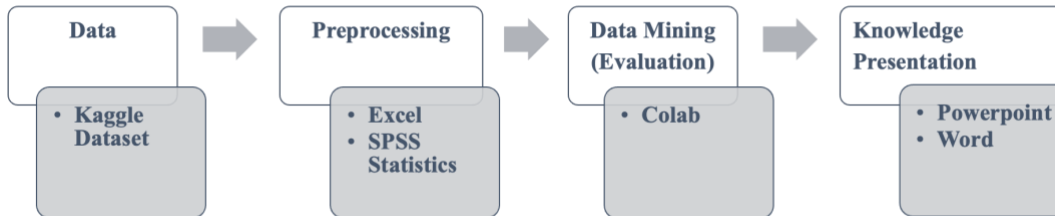
3.1.The Process Flow of Data Analysis

To utilize data mining techniques to discover intriguing and useful knowledge, data preprocessing is required. Although the cardiovascular dataset has no missing values, the measurement of data distribution is still essential. Thus, **SPSS Statistics** is an ideal instrument to evaluate the data distribution accurately. Additionally, the attributes ‘height’ and ‘weight’ are less practical because the two attributes are intertwined. Thus, the BMI score is a more comprehensive measurement of body condition.

On top of that, the attribute ‘age’ is in days, thus converting it to years is crucial. After finishing data preprocessing, the **Colab** is an ideal environment to construct numerous classifiers simultaneously. Finally, model evaluation and results visualization are implemented to express the discovered knowledge and patterns.

The project schedule can be summarized in Figure 1.

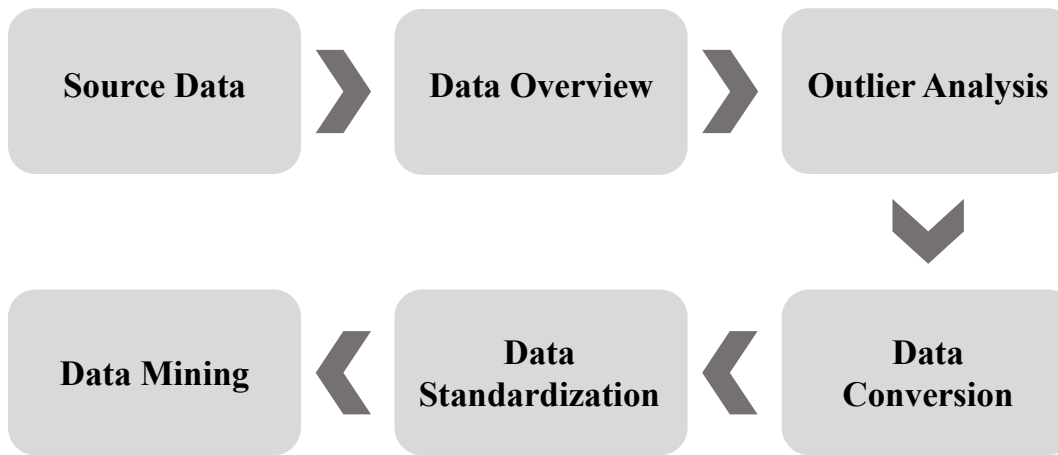
Figure 1. The project schedule



3.2.Data Preprocessing - Outline

As mentioned above, the goal of data preprocessing is to obtain high-quality data, which should possess characteristics, such as accuracy, completeness, consistency, interpretability, and reliability. To achieve the requirements, removing unnecessary and creating more interpretable columns/attributes and records/entities is essential. Meanwhile, outlier analysis is to measure whether extreme values exist or not and data discretization is to reduce the impact of fluctuation data. After finishing the preparation work, data mining is another work. In summary, the process flow of data preprocessing can be outlined in the subsequent Figure 2.

Figure 2. The process flow of data preprocessing



3.3.Data Overview

The SPSS statistics report can provide an understandable analysis for analysts. From the following reports (Figures 3 and 4), it is obvious that the cardio dataset contains no missing values, all valid numbers are 70,000. However, if we inspect Figure 3 more closely, there are some values beyond the normal situation. For example, the minimum values of ‘ap_hi’ and ‘ap_lo’ are minus, and the maximum of them both exceed 10,000, which is definitely abnormal. On top of that, from Figure 5 we know that the normal systolic blood pressure should be lower than 120mmHg and the diastolic blood pressure should also be below 80mmHg. When one of the values goes over the limit, it is considered elevated/high blood pressure. Therefore, adjusting/deleting some exceptional data is necessary.

Figure 3. SPSS report for numerical data(unprocessed)

		Statistics				
N	Valid	70000	70000	70000	70000	70000
	Missing	0	0	0	0	0
Mean		19468.87	164.36	74.2057	128.82	96.63
Std. Error of Mean		9.325	.031	.05441	.582	.712
Median		19703.00	165.00	72.0000	120.00	80.00
Mode		18236 ^a	165	65.00	120	80
Std. Deviation		2467.252	8.210	14.39576	154.011	188.473
Variance		6087330.790	67.406	207.238	23719.517	35521.895
Skewness		-.307	-.642	1.012	85.296	32.114
Std. Error of Skewness		.009	.009	.009	.009	.009
Kurtosis		-.823	7.944	2.587	7580.075	1425.915
Std. Error of Kurtosis		.019	.019	.019	.019	.019
Range		12915	195	190.00	16170	11070
Minimum		10798	55	10.00	-150	-70
Maximum		23713	250	200.00	16020	11000

Figure 4. SPSS report for categorical data(unprocessed)

		Statistics	
	N	Valid	Missing
		70000	0
gender		70000	0
cholesterol		70000	0
gluc		70000	0
smoke		70000	0
alco		70000	0
active		70000	0
cardio		70000	0

Figure 5. Blood pressure category by American Heart Association

BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)	and/or	DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
<u>HYPERTENSIVE CRISIS</u> (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Figure 6 includes seven frequency tables that display the distribution of categorical attributes. Notably, the target variable 'cardio' indicates an equal distribution between those with and without CVDs. The 'gender' attribute suggests that females make up nearly two times the proportion of males. The 'smoke' attribute shows that a small proportion of people smoke. In addition, the 'gluc' attribute suggests that less than 15% of people have high blood sugar, while the proportion of those with high cholesterol is less than 26%. The alcohol intake attribute shows that only a few people drink alcohol, while most people claim to perform physical activity.

Figure 6. The frequency table of ordinal and nominal attributes

gender			smoke			gluc			cholesterol		
	N	%		N	%		N	%		N	%
1	45530	65.0%	0	63831	91.2%	1	59479	85.0%	1	52385	74.8%
2	24470	35.0%	1	6169	8.8%	2	5190	7.4%	2	9549	13.6%
						3	5331	7.6%	3	8066	11.5%

alco			active			cardio		
	N	%		N	%		N	%
0	66236	94.6%	0	13739	19.6%	0	35021	50.0%
1	3764	5.4%	1	56261	80.4%	1	34979	50.0%

3.4.Outlier Analysis

When implementing outlier analysis, the boxplot and histogram can be utilized to evaluate possible outliers and offer accessible concepts. Usually, outlier analysis is performed on numerical data. Thus, the succeeding content will focus on the 'age', 'height', 'weight', 'ap_hi', and 'ap_lo' attributes. Speaking of boxplot, the

outliers can be defined as
$$\begin{cases} \text{Outlier} = Q_1 - 1.5IQR \\ \text{or} \\ \text{Outlier} = Q_3 + 1.5IQR \end{cases}$$
 . (see Figure 7) Note that $IQR = Q_3 - Q_1$.

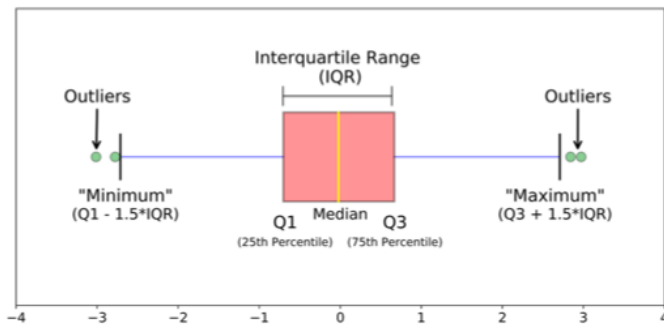


Figure 7. Boxplot

By examining the histogram and boxplot of the unprocessed 'age' attribute (Figures 8 and 9), the mean is 19468.87, and the standard deviation is 2467.252. Figures 8 and 9 show that the distribution is a little left-skewed. These phenomena suggest that there may exist some possible outliers and they will be further processed in the data conversion phase.

Figure 8. Unprocessed 'age' histogram

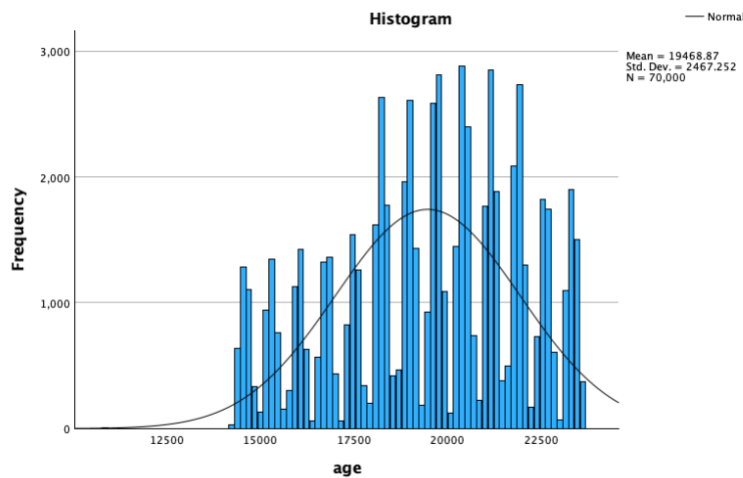
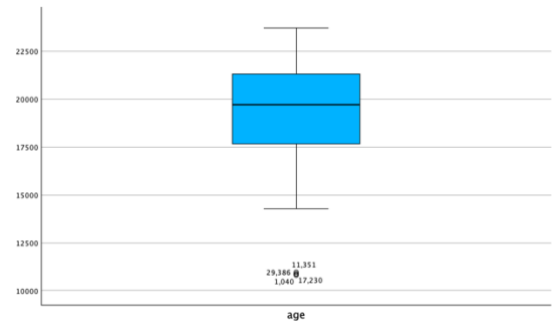


Figure 9. Unprocessed 'age' boxplot



By examining the histogram and boxplot of the unprocessed 'height' attribute (Figures 10 and 11), the mean is 164.36, and the standard deviation is 8.21. Figures 10 and 11 show that the distribution is left-skewed. At the same time, there are numerous possible outliers presented outside the upper and lower extremes. These phenomena suggest that outliers truly exist, and some adjustments must be implemented.

Figure 10. Unprocessed 'height' histogram

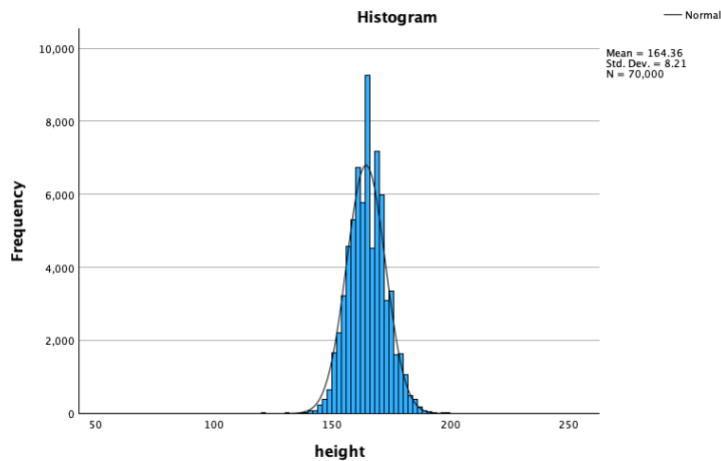
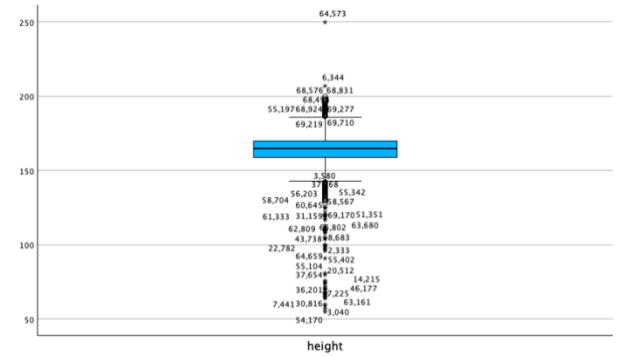


Figure 11. Unprocessed 'height' boxplot



By examining the histogram and boxplot of the unprocessed 'weight' attribute (Figures 12 and 13), the mean is 74.21, and the standard deviation is 14.396. Figures 12 and 13 show that the distribution is right-skewed. At the same time, there are numerous possible outliers presented outside the upper and lower extremes, the same as the 'height'. These phenomena suggest that outliers truly exist, and some adjustments must be implemented.

Figure 12. Unprocessed 'weight' histogram

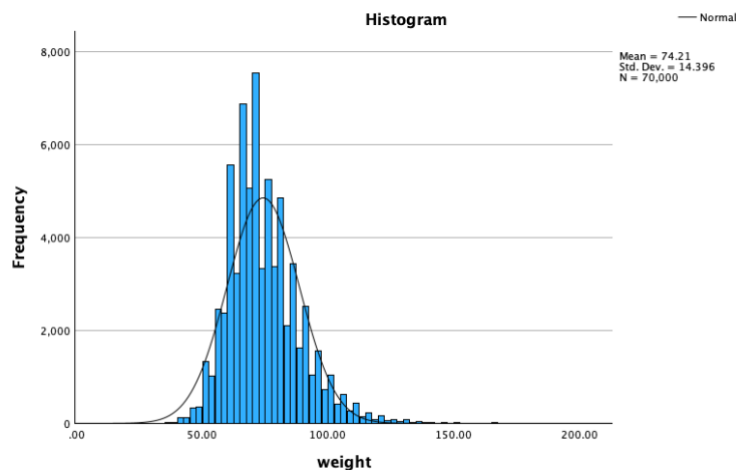
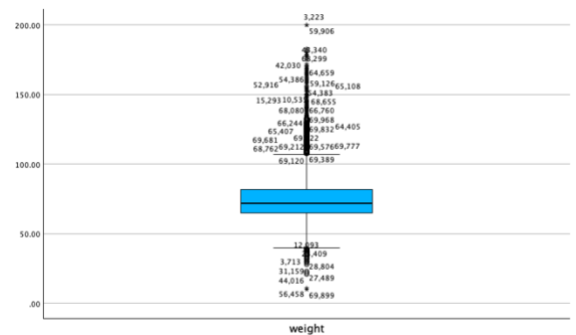


Figure 13. Unprocessed 'weight' boxplot



By examining the histogram and boxplot of the unprocessed 'ap_hi' and 'ap_lo' attributes altogether (Figures 14, 15, 16, and 17), the mean of 'ap_hi' is 128.82, and 'ap_lo' is 96.93. The standard deviation of 'ap_hi' is 154.011, and the 'ap_lo' is 188.473. Figures 14, 15, 16, and 17 show that the distribution of both attributes is right-skewed. At the same time, there are numerous possible outliers presented outside the upper extremes. These phenomena suggest that outliers truly exist, and some adjustments must be made.

Figure 14. Unprocessed 'ap_hi' histogram

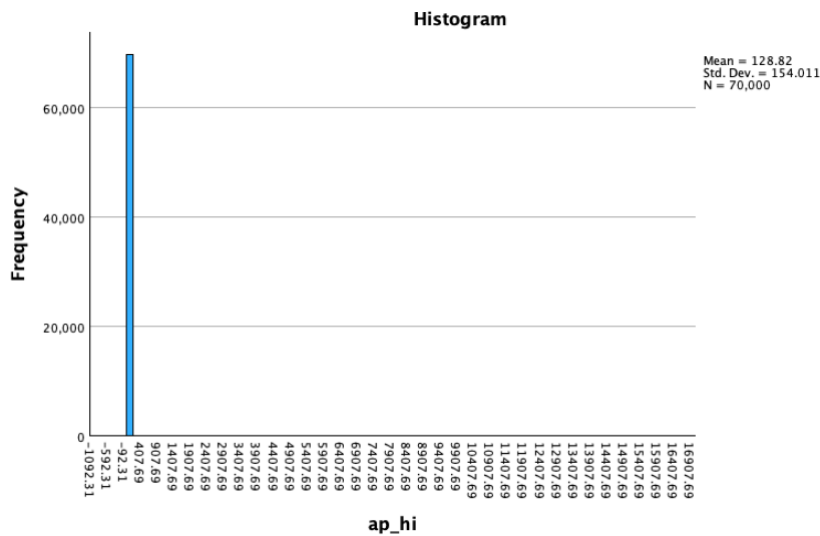


Figure 15. Unprocessed 'ap_hi' boxplot

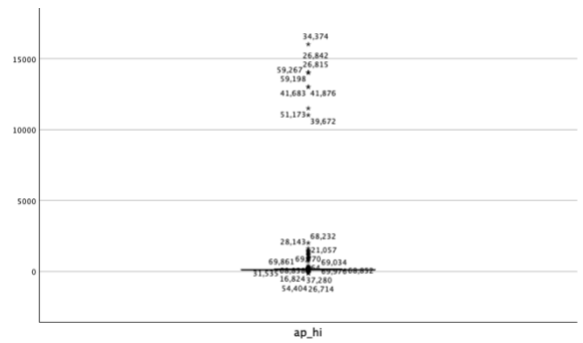


Figure 16. Unprocessed 'ap_lo' histogram

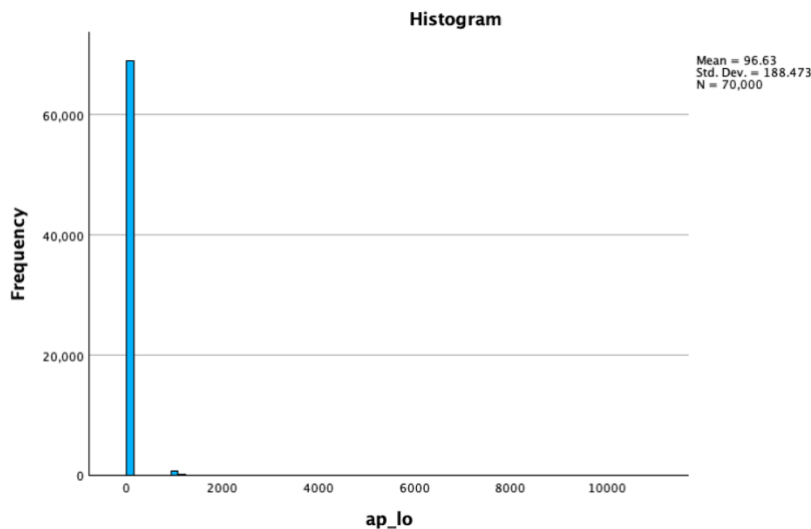
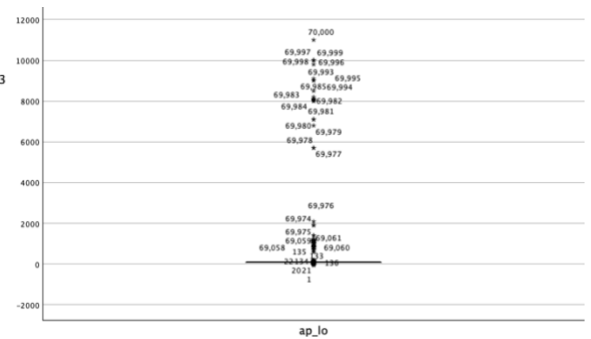


Figure 17. Unprocessed 'ap_lo' boxplot



When it comes to the process of skewness, square root, cube root, and logarithms are usually suitable for right-skewed distribution whereas square and cube are mainly for left-skewed distribution. Generally, box-cox is for both skewness. However, for convenience, the following distribution adjustment will use the log () method to deal with the skewness problem and try to reduce the effect of outliers, especially right-skewed. These procedures mainly cover 'height' and 'weight' attributes. For 'ap_hi' and 'ap_lo', deleting outliers directly can just deal with the problem due to the extreme values are all atypical.

After adjusting, the mean of 'height' reduces to 5.1, and the standard deviation is 0.45. From Figure 19 can find that there are no possible outliers anymore and the distribution is equally normal.

Figure 18. Processed ‘height’ histogram

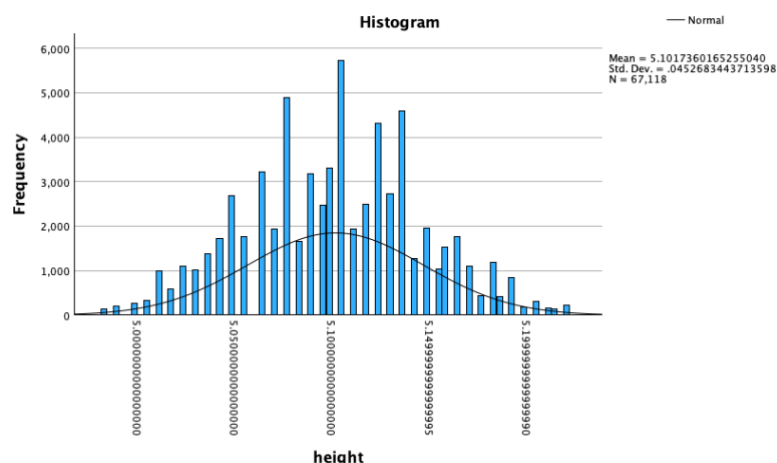
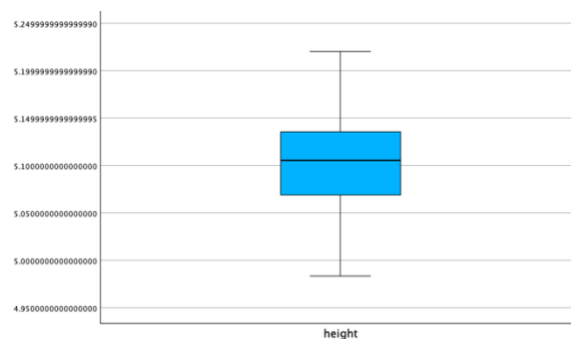


Figure 19. Processed ‘height’ boxplot



After adjusting, the shape of ‘weight’ is bell-shaped (see Figure 20) although there are some possible outliers presented on the upper extreme of Figure 21. The mean of it is nearly 4.29, and the standard deviation is approximately 1.75.

Figure 20. Processed ‘weight’ histogram

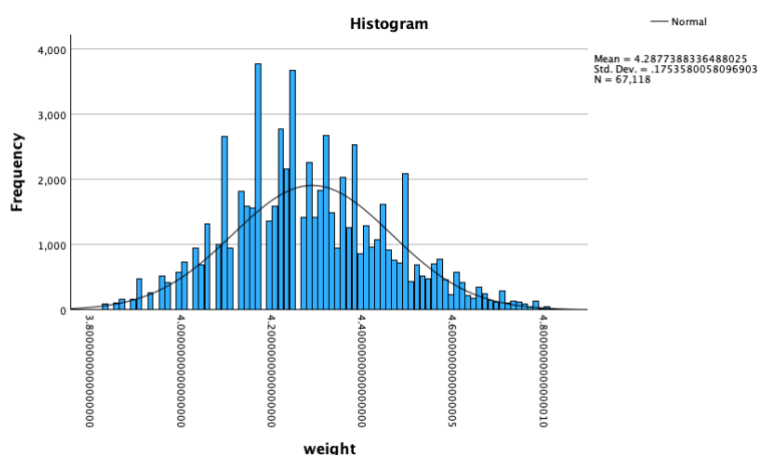
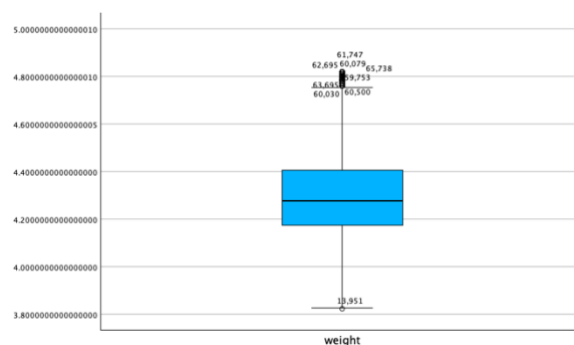


Figure 21. Processed ‘weight’ boxplot



For the situation of adjusting ‘ap_hi’ and ‘ap_lo’ (see Figures 22, 23, 24, and 25), it is obvious that the distribution of both attributes becomes nearly bell-shaped although both attributes still have some possible outliers located on the upper and lower extremes. The mean of ‘ap_hi’ is 126.65, and the standard deviation is 16.625, and the mean of ‘ap_lo’ is 81.25, and the standard deviation is 9.589.

Figure 22. Processed 'weight' histogram

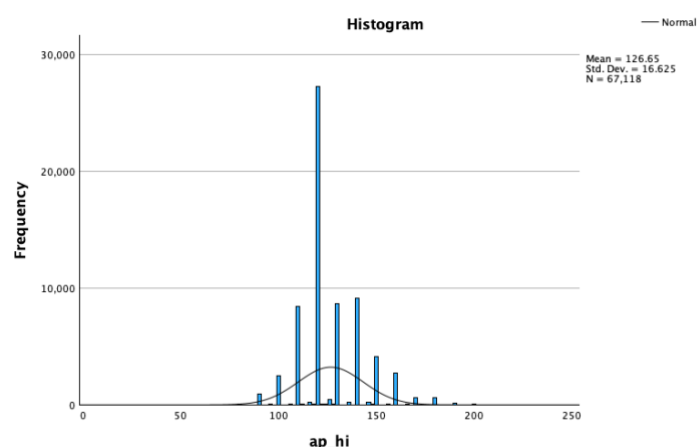


Figure 23. Processed 'weight' boxplot

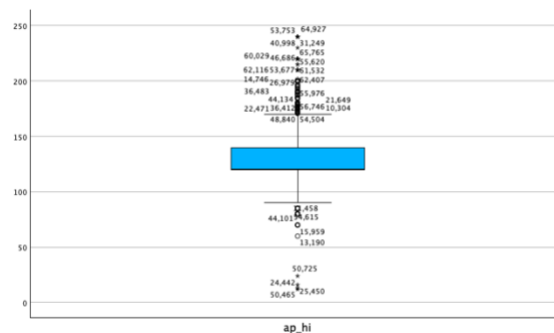


Figure 24. Processed 'weight' histogram

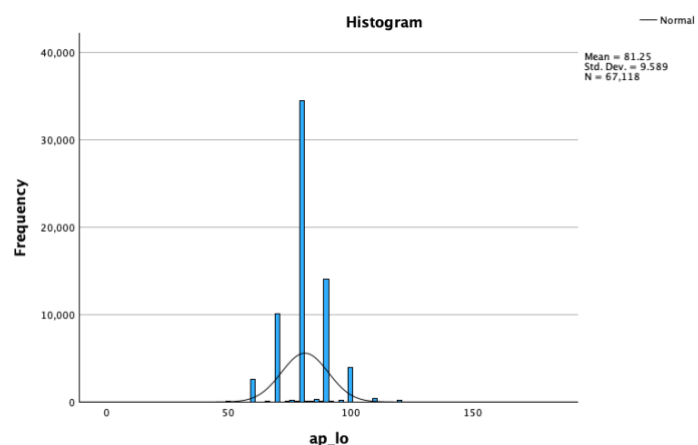
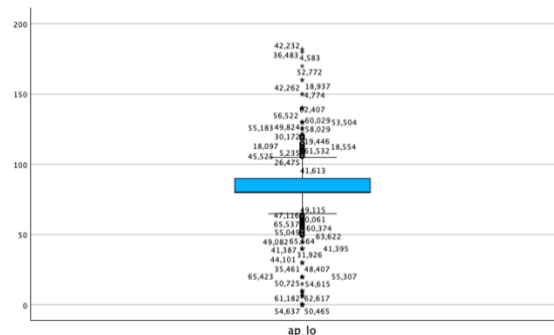


Figure 25. Processed 'weight' boxplot



In conclusion, for the numeric type of data, the first procedure is to look at their original distribution and then examine the reasonableness either utilizing `log ()` to adjust the outliers and skewness or remove those insane data directly.

3.5.Data Conversion

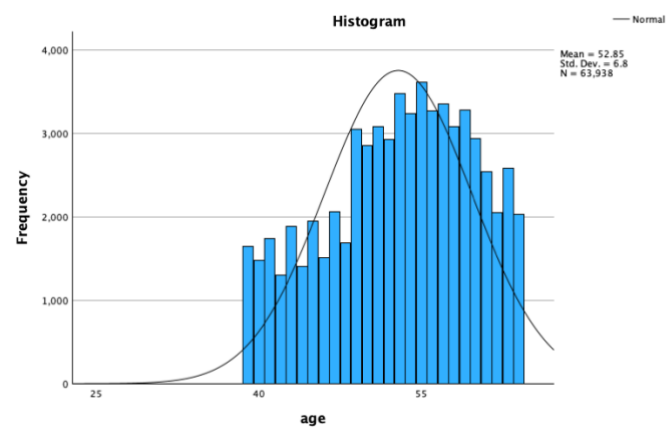
In the data conversion phase, 'age', 'height', and 'weight' attributes will be handled. At first, the unit of 'age' will be changed from 'days' to 'years' because that is more straightforward to analyze the factor. However, some inaccurate results may be obtained due to the days divided by 365 directly. These biases should be ignored because a little shift will not have a significant influence in detecting CVDs. In addition, a new 'BMI' attribute will be created since the body mass index (BMI) is a measurement that evaluates personal height and weight as a whole.

The following Figures 26 and 27 exhibit that the skewness and kurtosis of the converted 'age' attribute are acceptable because they are all under the limit of ± 1 . From the histogram, we still can speculate that there are possible outliers due to its shape and the minimum value in the descriptive report, which is 29 whereas the mean is 52.85.

Figure 26. Processed ‘age’ descriptive

Descriptives			
		Statistic	Std. Error
age	Mean	52.85	.027
	95% Confidence Interval for Mean	Lower Bound	52.80
		Upper Bound	52.91
	5% Trimmed Mean	53.00	
	Median	53.00	
	Variance	46.246	
	Std. Deviation	6.800	
	Minimum	29	
	Maximum	64	
	Range	35	
	Interquartile Range	10	
	Skewness	-.303	.010
	Kurtosis	-.837	.019

Figure 27. Processed ‘age’ histogram

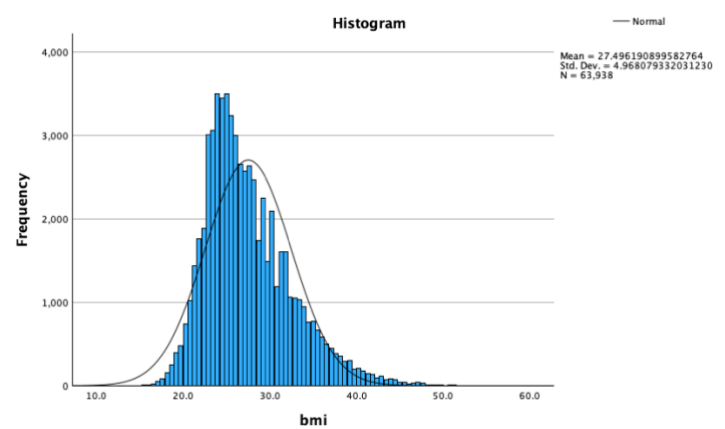


The following Figures 28 and 29 exhibit that the skewness and kurtosis of the generated ‘bmi’ attribute contain some anomaly due to the kurtosis being beyond 1, however, it could be overlooked since the sample (N) is huge and the value isn’t dramatic. From the histogram, the distribution of it is nearly bell-shaped, therefore it can undergo further processing, that is, data standardization.

Figure 28. Generated ‘bmi’ descriptive

Descriptives			
		Statistic	Std. Error
bmi	Mean	27.4961909	.019647577
	95% Confidence Interval for Mean	Lower Bound	27.4576816
		Upper Bound	27.5347002
	5% Trimmed Mean	27.2138084	
	Median	26.5432099	
	Variance	24.682	
	Std. Deviation	4.96807933	
	Minimum	14.6092038	
	Maximum	56.2957403	
	Range	41.6865365	
	Interquartile Range	6.49191441	
	Skewness	.936	.010
	Kurtosis	1.049	.019

Figure 29. Generated ‘bmi’ histogram

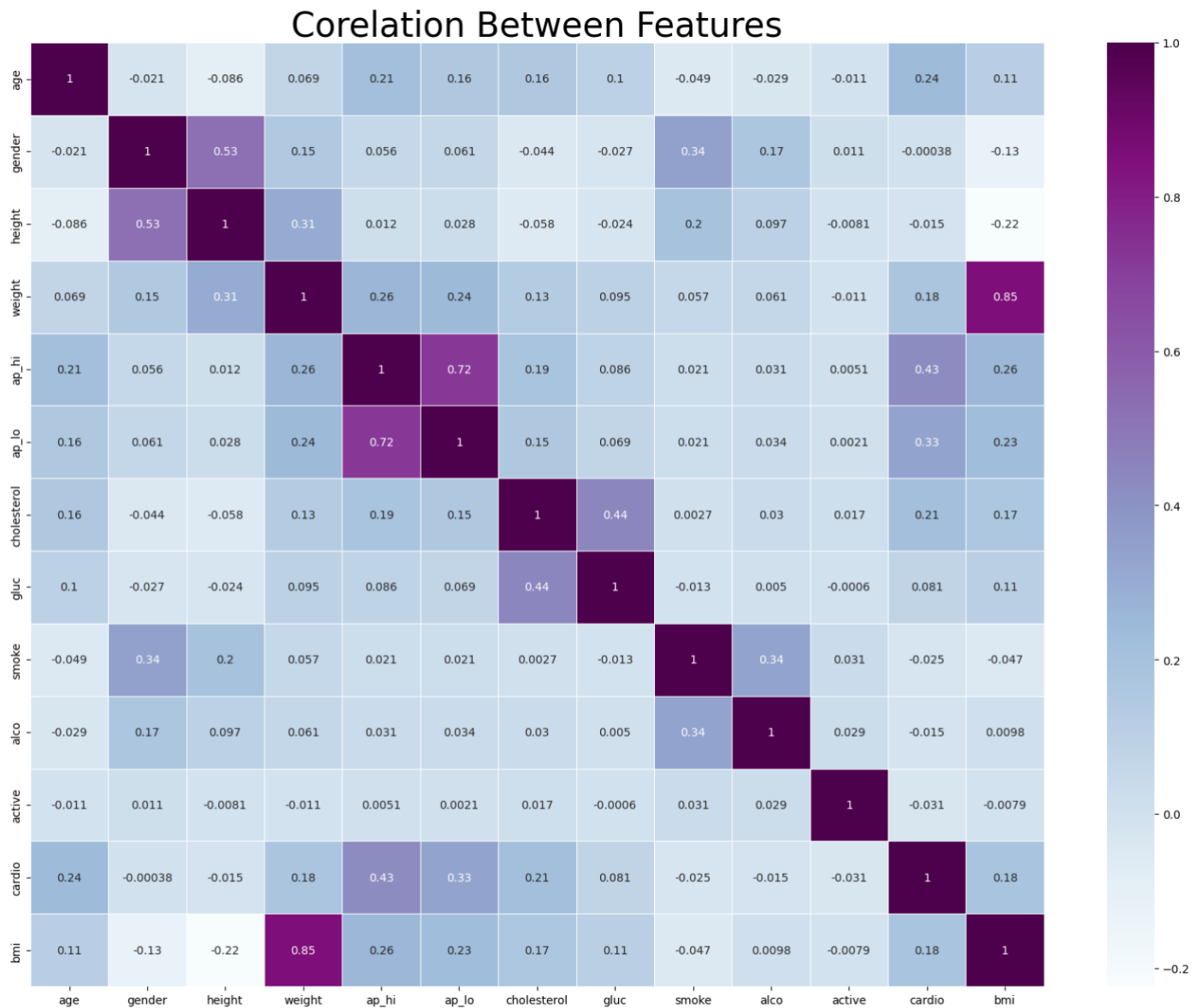


3.6.Data Standardization

Before undergoing data standardization, the heat map is a great tool to visualize the correlation between the target value and the input values. By using it, the final input features can be determined, which have a relatively strong correlation to the target value. (Figure 30)

From Figure 30, there is no doubt that the ‘age’, ‘ap_hi’, ‘ap_lo’, and ‘cholesterol’ have comparatively low positive correlations. Although the normal definition of having a low positive correlation should be greater than 0.3 based on the rule of thumb, however, due to the property of the data, the criterion should be more loosely. Thus, only input features with a correlation value greater than 0.2 should be considered.

Figure 30. Correlation between attributes



After selecting the final input features, the StandardScaler () method is applied, which standardizes features by removing the mean and scaling to unit variance. It can achieve mean removal and variance scaling, which means each input feature has a mean of 0 and scales these features to have a standard deviation of 1.

4. The Usage of Data Mining Approaches and Performance Evaluation

4.1.The Usage of Data Mining Approaches

Classification algorithms are implemented in this final project. In classification, k-nearest neighbors (k-NN), random forest, support vector machine (SVM), and XGBoost methods are expected to be constructed to serve the purpose of developing a predictive model that can assist in making decisions or predictions related to cardiovascular disease. During training, different hyperparameters will be tried, then to find the best combination. For the final input feature, which is ‘age’, ‘ap_hi’, ‘ap_lo’, and ‘cholesterol’, respectively. The

output feature is whether the outcome has CVDs or not, that is a binary classification. Moreover, the training and testing split is 75%-25%, as shown in the following code:

```
X_train, X_test, y_train, y_test = train_test_split(standard_X, Y, test_size=0.25,
random_state=42, shuffle = True)
```

About the k-NN algorithm, it is easy to understand and implement. The fundamental concept of k-NN involves identifying the nearest neighbors to a new input instance and leveraging their labels or values to formulate predictions. It also can perform well with complex and non-linear decision boundaries and can adapt to different types of data distributions although the increased amount of data can lead to high computational complexity and lengthy prediction time. Nevertheless, the k-NN is used to predict the CVDs.

Table 2. Training process of k-NN

```
params = {'n_neighbors':list(range(0, 50)),
          'weights':['uniform', 'distance'],
          'p':[1,2]}

knn = KNeighborsClassifier()

knn_grid_cv = GridSearchCV(knn, param_grid=params, cv=10)

knn_grid_cv.fit(X_train, y_train)

print("Best Hyper Parameters:\n",knn_grid_cv.best_params_)
```

After training, the best combination of hyperparameters is {'n_neighbors': 42, 'weights': 'uniform', 'p': 1}, which means the number of neighbors used is 42, using Manhattan distance, and all points in each neighborhood are weighted equally.

As for the random forest algorithm, it is an ensemble learning method that can be used for both classification and regression tasks. Comprising a collection of decision trees, each tree in the ensemble is constructed from a data sample drawn from the training set with replacement, known as the bootstrap sample. Within this training sample, one-third is designated as test data, referred to as the out-of-bag (OOB) sample. Another layer of randomness is introduced through feature bagging, enhancing dataset diversity, and diminishing correlation among the decision trees. The prediction strategy varies depending on the problem type. For this project, classification will be performed, thus the prediction is determined by a majority vote (i.e., the most frequent categorical variable). In addition, it is suitable for large datasets has high accuracy, and is robust to overfitting, however, a larger number of trees will cause high computational complexity.

Table 3. Training process of random forest

```
params = {  
    'n_estimators': [10, 50, 100, 150, 200, 300, 400, 500],  
    'max_depth' : [10,20,30,40,50],  
    'criterion' : ['entropy','gini']  
}  
  
rfc_gridcv = RandomForestClassifier(random_state=42)  
  
rfc_gridcv = GridSearchCV(estimator=rfc_gridcv, param_grid=params, cv= 10, n_jobs = -  
1)  
  
rfc_gridcv.fit(X_train, y_train)  
  
print("Best Hyper Parameters:\n",rfc_gridcv.best_params_)
```

After completion of the training process, it has been determined that the optimal combination of hyperparameters has been identified. {'n_estimators': 500, 'max_depth': 10, 'criterion': 'entropy'}, which represents the number of trees is 500, the maximum depth of the tree is 10, and the Gini impurity to measure the quality of the split.

Regarding SVM, it can be effective in high-dimensional spaces and has different kernel functions. SVM operates by transforming data into a high-dimensional feature space, enabling the categorization of data points, even in cases where they are not easily linearly separable. The algorithm identifies a separator between the categories, and through a transformation of the data, this separator can be represented as a hyperplane. Subsequently, the features of new data can be analyzed to predict the group to which a new record should be assigned. In summary, it is a machine learning algorithm that is capable of dealing with global optimality through hyperplane and being robust to overfitting. However, it should be noted that SVM is sensitive to noisy data and outliers in the dataset, which can significantly impact its performance. Moreover, the training process could be lengthy when dealing with large datasets, and the choice of kernel is also crucial.

Table 4. Training process of SVM

```
param_grid = {  
    'C': [0.1, 1, 10],  
    'kernel': ['poly', 'rbf'],  
    'gamma': ['scale', 'auto'],  
}  
  
svm_classifier = SVC(random_state=42)  
  
grid_search = GridSearchCV(svm_classifier, param_grid, cv=10)
```

```
grid_search.fit(X_train, y_train)

print("Best Hyper Parameters:\n",grid_search.best_params_)
```

After the completion of training, the best hyperparameters can be obtained {'C': 1, 'kernel': 'rbf', 'gamma': 'scale'}, which illustrates the regularization parameter is 1, the type of kernel is rbf, and the kernel coefficient is scale.

Finally, XGBoost is utilized. It operates by amalgamating numerous weak learners to create a robust learner. A weak learner refers to a machine learning model that performs marginally better than random guessing. However, the collective effect of combining these weak learners results in a strong learner with significantly enhanced accuracy. XGBoost is known for its high performance and efficiency and for employing regularization mechanisms to prevent overfitting. However, it could be resource-intensive to cope with large datasets.

Table 5. Training process of XGBoost

```
xgb_params = {
    'learning_rate': [0.1, 0.2],
    'n_estimators': [100, 200, 300, 400, 500],
    'max_depth': [3, 5, 7],
    'colsample_bytree': [0.8, 1.0],
    'gamma': [0, 1, 5],
}

xgb_classifier = xgb.XGBClassifier(random_state=42)

xgb_grid_cv = GridSearchCV(xgb_classifier, param_grid=xgb_params, cv=10)

xgb_grid_cv.fit(X_train, y_train)

print("Best Hyper Parameters:\n",xgb_grid_cv.best_params_)
```

After training, the best hyperparameters are discovered {'learning_rate': 0.1, 'n_estimators': 300, 'max_depth': 3, 'colsample_bytree': 0.8, 'gamma': 0}, which stand for 0.1 learning rate used in the update, the number of trees is 300, maximum depth of a tree is 3, the subsample ratio of columns is 0.8, and the minimum loss reduction is 0.

4.2.Performance Evaluation

To acquire more accuracy than obtained from training, 10-fold cross-validation is implemented. In addition, accuracy, sensitivity, specificity, precision, and F1 score are examined. The formula of each criterion and confusion matrix are shown in the following Table 6, and the results are in Table 7.

Actual Values

Table 6. Confusion Matrix

Predicted Values	Positive	Positive	Negative
	Negative	Numbers of Cases Predicted with CVDs and Actually had CVDs (TP)	Numbers of Cases Predicted with CVDs and Actually had No CVDs (FP)
		Numbers of Cases Predicted with No CVDs and Actually had CVDs (FN)	Numbers of Cases Predicted with No CVDs and Actually had No CVDs (TN)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \text{ gives an overall view of correct predictions}$$

$$Sensitivity(Recall) = \frac{TP}{TP + FN}, \text{ the ability to correctly identify positive instances}$$

$$Specificity = \frac{TN}{TN + FP}, \text{ the ability to correctly identify negative instances}$$

$$Precision = \frac{TP}{TP + FP}, \text{ the accuracy of positive predictions}$$

$$F1 \text{ Score} = \frac{2 \times (Precision \times Sensitivity)}{(Precision + Sensitivity)}, \text{ balances precision and sensitivity}$$

Table 7. Experiment Result

	k-NN	Random Forest	SVM	XGBoost
True Positive	5583	5569	5448	5689
True Negative	6008	6025	6132	5905
False Positive	1891	1874	1767	1994
False Negative	2503	2517	2638	2397
Average Accuracy	0.7257	0.7278	0.7277	0.7293
Sensitivity	0.6905	0.6887	0.6738	0.7036
Specificity	0.7606	0.7628	0.7763	0.7476
Precision	0.746	0.7482	0.755	0.7405
F1 Score	0.71711	0.71721	0.71209	0.72157

From the above Table 7, it is clear that XGBoost outperforms three other algorithms in average accuracy, sensitivity, and F1 score index whereas SVM has the best performance considering specificity and precision. Due to the characteristics of the final project, the sensitivity, specificity, precision, and F1 score are thought to be more important. In medical diagnosis, the sensitivity is more critical than the others. Therefore, XGBoost has a relatively better performance.

5. Discovered Knowledge and Conclusion

In general, the main purpose of the project is to discover knowledge from a cardiovascular dataset and hope to gain various insights that contribute to a better understanding of factors influencing cardiovascular health. The objectives of the project are to identify risk factors and construct a predictive model. These models can be used as early detection indicators to assist patients and physicians in diagnosing diseases.

From the preprocessing stage, the heat map shows the correlation between each factor. Among them, the 'age', 'ap_hi', 'ap_lo', and 'cholesterol' attributes have low positive correlations. However, the 'BMI' factor's correlation is low, which surprises me. Then, utilizing these factors to construct four predictive models, which are k-NN, random forest, SVM, and XGBoost. From Table, XGBoost emerged as the standout performer, exhibiting superior sensitivity and F1 score. This robust performance positions XGBoost as the chosen classification and predictive model.

In conclusion, this project has successfully delved into a comprehensive analysis of a cardiovascular dataset, aiming to unravel crucial insights into the factors influencing cardiovascular health. Through meticulous preprocessing, the correlation analysis highlighted intriguing patterns, with attributes like 'age,' 'ap_hi,' 'ap_lo,' and 'cholesterol' demonstrating correlations. The implementation of XGBoost holds great promise in serving as an early detection tool, aiding both patients and physicians in diagnosing cardiovascular diseases promptly.

In essence, this project not only sheds light on the interplay of various factors affecting cardiovascular health but also provides a practical and effective tool in the form of the XGBoost predictive model. The findings contribute significantly to the ongoing efforts to enhance our understanding of cardiovascular risk factors and empower healthcare professionals in delivering proactive and personalized care.

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